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TECHNISCHE UNIVERSITÄT DARMSTADT

Is Synthetic Data all We Need? Benchmarking the Robustness of Models Trained with Synthetic Images


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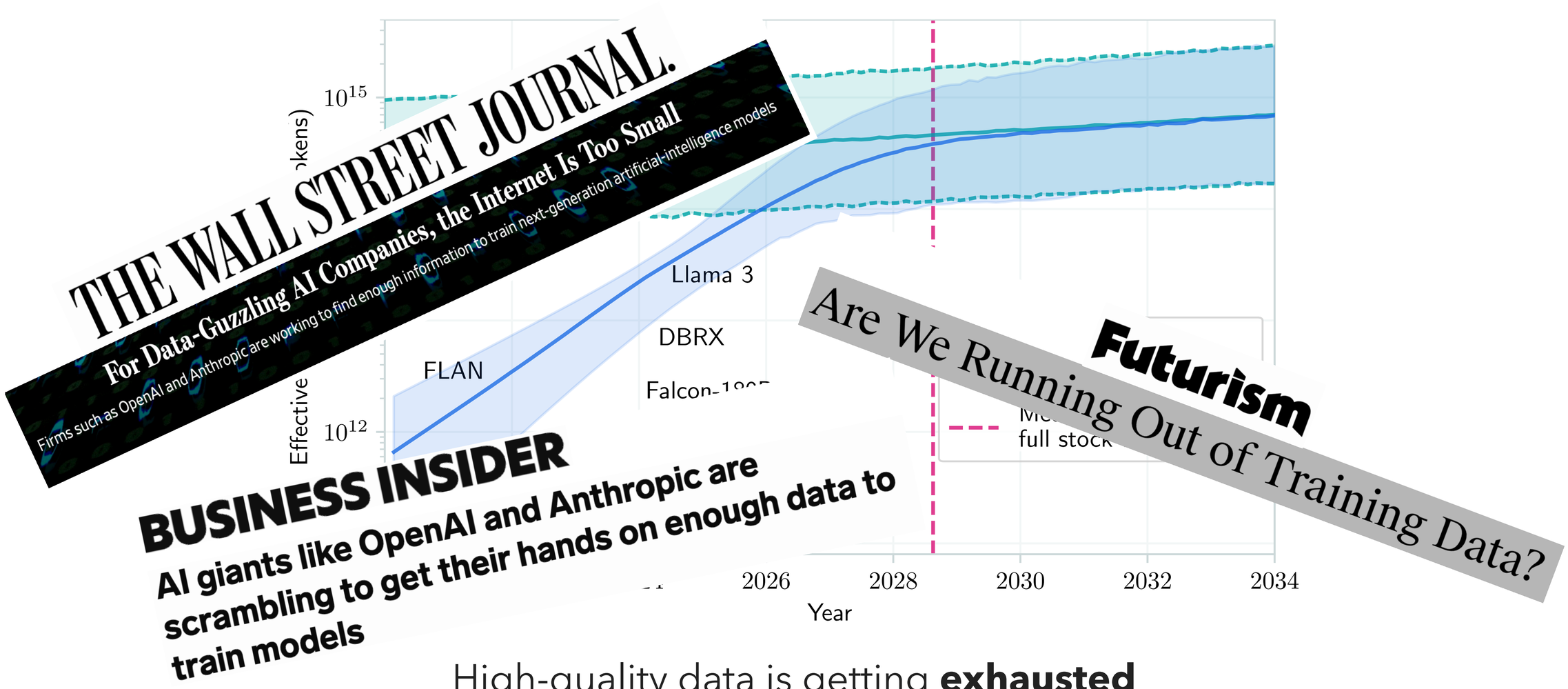

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Simone Schaub-Meyer[1,2]


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Modern models are data-hungry

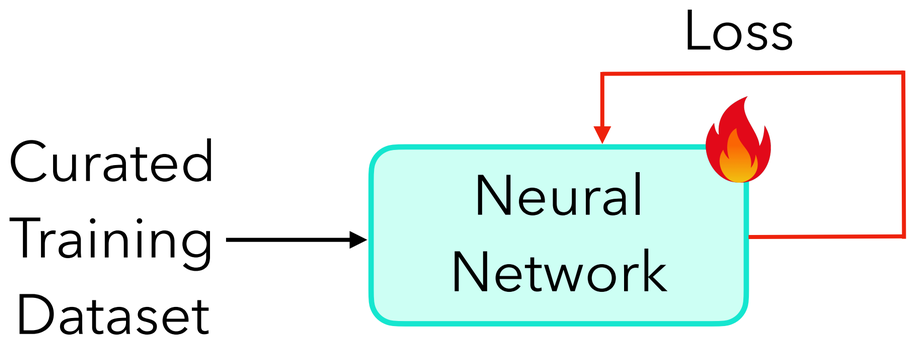


High-quality data is getting **exhausted**

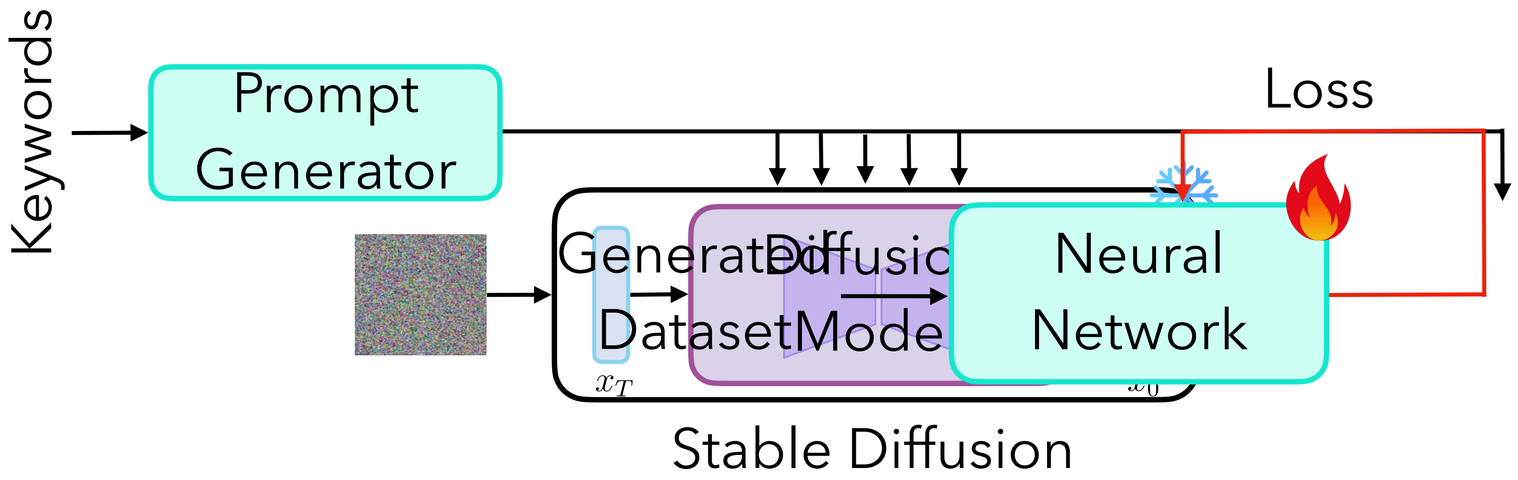
Image Credit: Stable Diffusion XL

[1] Villalobos *et al.,* Will we run out of data? An analysis of the limits of scaling datasets in machine learning, *arXiv:2211.04325* (2022).

Traditional training pipeline



Generated data to the rescue



Synthetic Clones: Any model trained using generated data from a foundational model.

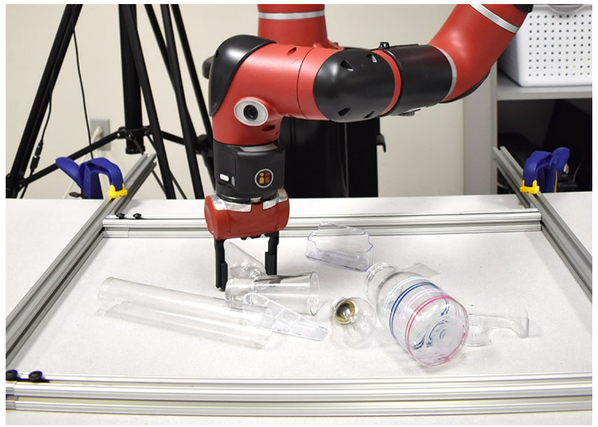
Importance of robustness

Autonomous driving



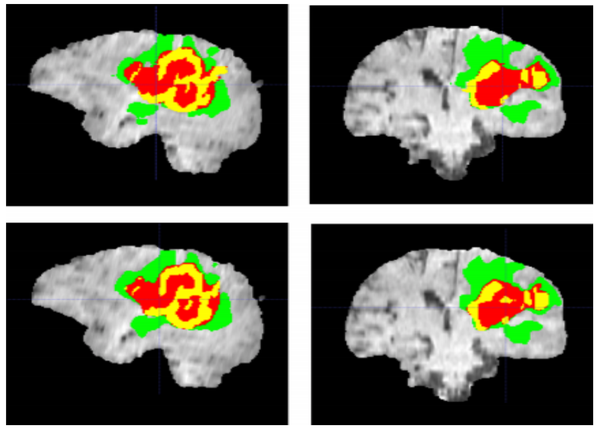
Robust to weather, background, etc.

Robotics



OOD detection

Medical imaging



Calibration

Cordts *et al*., The Cityscapes dataset for semantic urban scene understanding, In CVPR 2016.

Importance of robustness

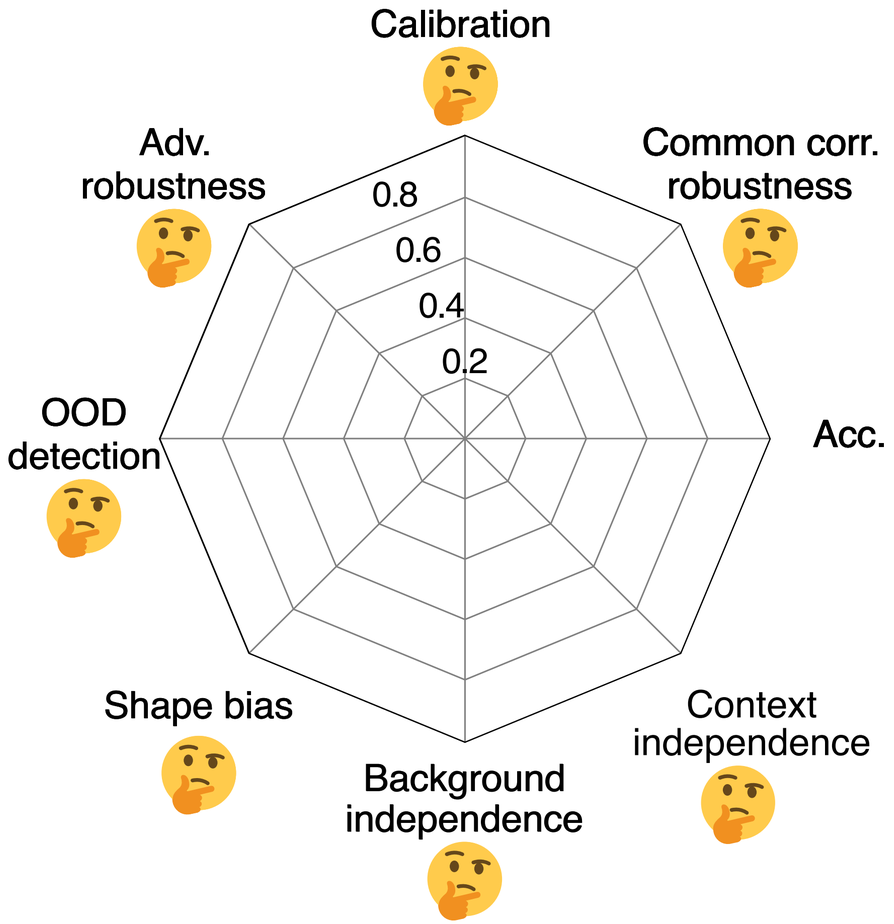


Our work

We benchmark synthetic clones for multiple robustness measures

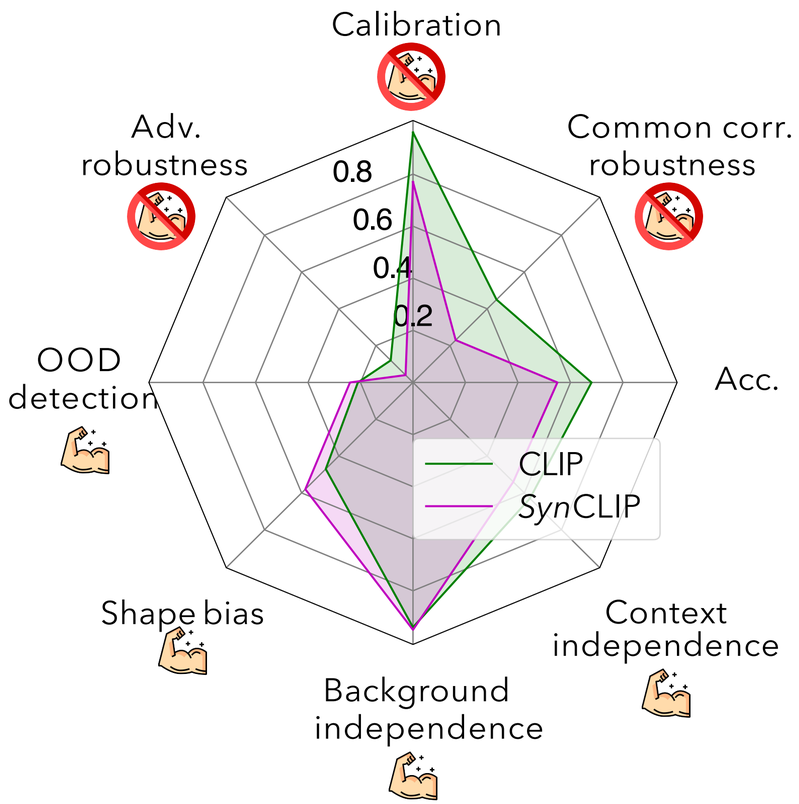
across multiple categories of models.

🤔: Unknown

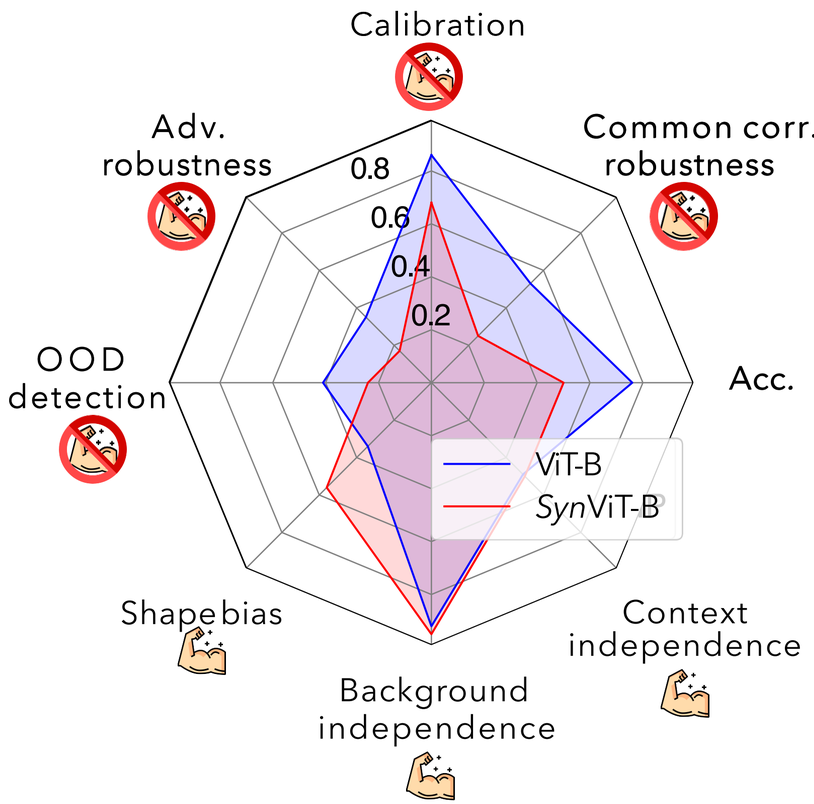


- Supervised ***Syn***Clone
- Self-Supervised ***Syn***Clone
- Multimodal ***Syn***Clone

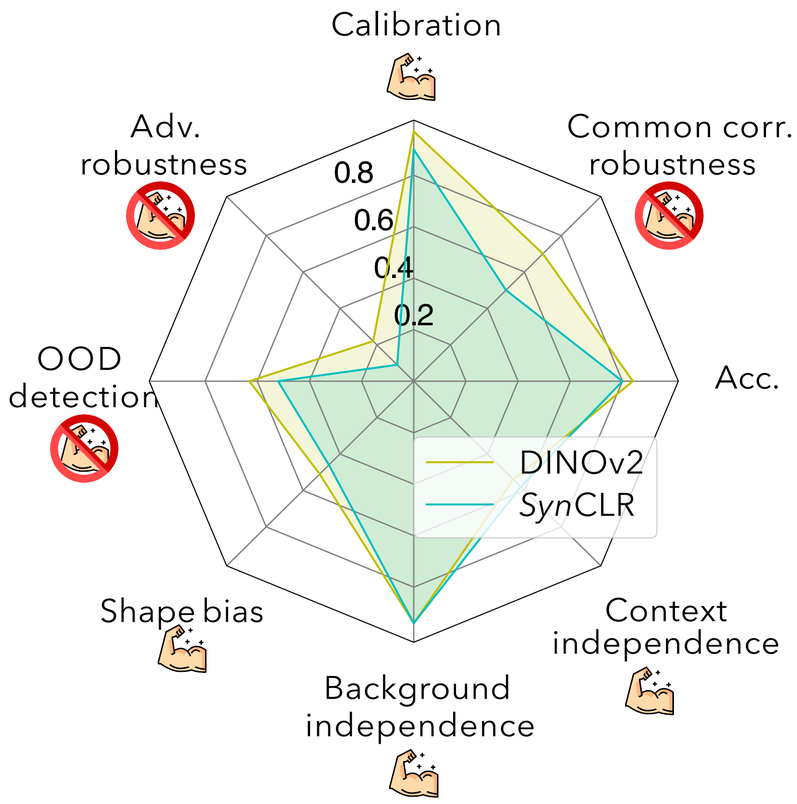
Is synthetic data all we need?



Multimodal Synthetic Clones



Supervised Synthetic Clones



SSL Synthetic Clones

Is synthetic data all we need?

Well, not quite yet…

Adversarial perturbations



Common corruptions



OOD detection



Calibration



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Acknowledgements:

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See you at the poster session at 11:40 am